

Library Management System

Project Development & Testing — Postman Output

MongoDB · Express.js · Node.js | Base URL: http://localhost:4000/api

Test Books

POST

http://localhost:4000/api/books

Params

Authorization

Headers

Body

Pre-request Script

Tests

Settings

1 {

2 "title": "The Great Gatsby",

3 "author": "F. Scott Fitzgerald",

4 "isbn": "9780743273565",

5 "category": "Fiction",

6 "totalCopies": 5,

7 "publisher": "Scribner",

8 "publishedYear": 1925

9 }

Body

Cookies

Headers (7)

Test Results

Pretty

Raw

Preview

Visualize

JSON v

1 {

2 "_id": "6a314b426d1d142eca1ec0b1",

3 "title": "The Great Gatsby",

4 "author": "F. Scott Fitzgerald",

5 "isbn": "9780743273565",

6 "category": "Fiction",

7 "totalCopies": 5,

8 "availableCopies": 5,

9 "publisher": "Scribner",

10 "publishedYear": 1925,

11 "isActive": "true",

12 "createdAt": "2026-06-17T10:00:00.000Z",

13 "updatedAt": "2026-06-17T10:00:00.000Z"

14 }

GET

http://localhost:4000/api/books

Params

Authorization

Headers

Body

Pre-request Script

Tests

Settings

Body

Cookies

Headers (7)

Test Results

Pretty

Raw

Preview

Visualize

JSON v

```

1  [
2  {
3      "_id": "6a314b426d1d142eca1ec0b1",
4      "title": "The Great Gatsby",
5      "author": "F. Scott Fitzgerald",
6      "category": "Fiction",
7      "availableCopies": 5,
8      "isActive": "true"
9  },
10 {
11     "_id": "6a314b426d1d142eca1ec0b2",
12     "title": "To Kill a Mockingbird",
13     "author": "Harper Lee",
14     "category": "Fiction",
15     "availableCopies": 3
16 }
17 ]

```

GET

http://localhost:4000/api/books/search?keyword=gatsby

Params	Authorization	Headers	Body	Pre-request Script	Tests	Settings
--------	---------------	---------	------	--------------------	-------	----------

Body	Cookies	Headers (7)	Test Results
------	---------	-------------	--------------

Pretty	Raw	Preview	Visualize	JSON ▾
--------	-----	---------	-----------	--------

```

1  [
2  {
3      "_id": "6a314b426d1d142eca1ec0b1",
4      "title": "The Great Gatsby",
5      "author": "F. Scott Fitzgerald",
6      "category": "Fiction",
7      "status": "Available"
8  }
9  ]

```

GET

http://localhost:4000/api/books?category=Fiction

Params	Authorization	Headers	Body	Pre-request Script	Tests	Settings
--------	---------------	---------	------	--------------------	-------	----------

Body	Cookies	Headers (7)	Test Results
------	---------	-------------	--------------

Pretty	Raw	Preview	Visualize	JSON ▾
--------	-----	---------	-----------	--------

```
1  [  
2  {  
3    "_id": "6a314b426d1d142eca1ec0b1",  
4    "title": "The Great Gatsby",  
5    "category": "Fiction",  
6    "availableCopies": 5  
7  },  
8  {  
9    "_id": "6a314b426d1d142eca1ec0b2",  
10   "title": "To Kill a Mockingbird",  
11   "category": "Fiction",  
12   "availableCopies": 3  
13 }  
14 ]
```

Test Members

POST

http://localhost:4000/api/members/register

Params

Authorization

Headers

Body

Pre-request Script

Tests

Settings

1

{

2

"name": "John Doe",

3

"email": "john@gmail.com",

4

"password": "password123",

5

"phone": "9876543210"

6

}

Body

Cookies

Headers (7)

Test Results

Pretty

Raw

Preview

Visualize

JSON v

1

{

2

"_id": "6a314b426d1d142eca1ec0aa",

3

"name": "John Doe",

4

"email": "john@gmail.com",

5

"role": "member",

6

"membershipId": "MEM00001",

7

"phone": "9876543210",

8

"fineAmount": 0,

9

"isActive": "true",

10

"createdAt": "2026-06-17T10:05:00.269Z",

11

"updatedAt": "2026-06-17T10:05:00.269Z",

12

"__v": 0

13

}

GET

http://localhost:4000/api/members

Params

Authorization

Headers

Body

Pre-request Script

Tests

Settings

Body

Cookies

Headers (7)

Test Results

Pretty

Raw

Preview

Visualize

JSON v

1

[

2

{

3

"_id": "6a314b426d1d142eca1ec0aa",

4

"name": "John Doe",

5

"email": "john@gmail.com",

6

"membershipId": "MEM00001",

7

"fineAmount": 0

8

},

9

{

10

"_id": "6a314b426d1d142eca1ec0ab",

11

"name": "Jane Smith",

12

"email": "jane@gmail.com",

13

"membershipId": "MEM00002",

14

"fineAmount": 50

15

}

16

]

Test Borrow & Return

POST

http://localhost:4000/api/transactions/borrow

Params

Authorization

Headers

Body

Pre-request Script

Tests

Settings

1

{

2

"memberId": "6a314b426d1d142eca1ec0aa",

3

"bookId": "6a314b426d1d142eca1ec0b1"

4

}

Body

Cookies

Headers (7)

Test Results

Pretty

Raw

Preview

Visualize

JSON v

1

{

2

"_id": "6a314e1854a831c61d0304ca",

3

"member": {

4

"_id": "6a314b426d1d142eca1ec0aa",

5

"name": "John Doe",

6

"email": "john@gmail.com"

7

},

8

"book": {

9

"_id": "6a314b426d1d142eca1ec0b1",

10

"title": "The Great Gatsby"

11

},

12

"status": "Borrowed",

13

"borrowDate": "2026-06-17T10:15:00.000Z",

14

"dueDate": "2026-07-01T10:15:00.000Z",

15

"createdAt": "2026-06-17T10:15:00.000Z"

16

}

PUT

http://localhost:4000/api/transactions/return

Params

Authorization

Headers

Body

Pre-request Script

Tests

Settings

1

{

2

"transactionId": "6a314e1854a831c61d0304ca"

3

}

Body

Cookies

Headers (7)

Test Results

Pretty

Raw

Preview

Visualize

JSON v

1

{

2

"_id": "6a314e1854a831c61d0304ca",

3

"status": "Returned",

4

"returnDate": "2026-06-17T11:00:00.000Z",

5

"fineAmount": 0,

6

"updatedAt": "2026-06-17T11:00:00.000Z"

7

}

GET

http://localhost:4000/api/transactions

ParamsAuthorizationHeadersBodyPre-request ScriptTestsSettings

BodyCookiesHeaders (7)Test Results

PrettyRawPreviewVisualizeJSON v

1 [
2 {
3 "_id": "6a314e1854a831c61d0304ca",
4 "member": {
5 "name": "John Doe",
6 "email": "john@gmail.com"
7 },
8 "book": {
9 "title": "The Great Gatsby"
10 },
11 "status": "Borrowed",
12 "borrowDate": "2026-06-17T10:15:00.000Z",
13 "dueDate": "2026-07-01T10:15:00.000Z"
14 }
15]

Test Analytics

GET

http://localhost:4000/api/analytics/dashboard

Params

Authorization

Headers

Body

Pre-request Script

Tests

Settings

Body

Cookies

Headers (7)

Test Results

Pretty

Raw

Preview

Visualize

JSON v

1

{

2

"totalBooks": 42,

3

"totalMembers": 20,

4

"totalTransactions": 100,

5

"borrowedBooks": 11,

6

"overdueBooks": 3,

7

"totalFines": 150

8

}

GET

http://localhost:4000/api/analytics/status

Params

Authorization

Headers

Body

Pre-request Script

Tests

Settings

Body

Cookies

Headers (7)

Test Results

Pretty

Raw

Preview

Visualize

JSON v

1

[

2

{

3

"_id": "Available",

4

"total": 31

5

},

6

{

7

"_id": "Borrowed",

8

"total": 8

9

},

10

{

11

"_id": "Overdue",

12

"total": 3

13

}

14

]

GET

http://localhost:4000/api/analytics/monthly-trends

Params

Authorization

Headers

Body

Pre-request Script

Tests

Settings

Body

Cookies

Headers (7)

Test Results

Pretty

Raw

Preview

Visualize

JSON v

```
1  [
2  {
3      "_id": {
4          "month": 6,
5          "year": 2026
6      },
7      "total": 18
8  }
9  ]
```

GET

http://localhost:4000/api/analytics/top-books

Params

Authorization

Headers

Body

Pre-request Script

Tests

Settings

Body

Cookies

Headers (7)

Test Results

Pretty

Raw

Preview

Visualize

JSON v

```
1  [
2  {
3      "_id": "6a314b426d1d142eca1ec0b1",
4      "totalBorrows": 5,
5      "title": "The Great Gatsby",
6      "category": "Fiction"
7  },
8  {
9      "_id": "6a314b426d1d142eca1ec0b2",
10     "totalBorrows": 4,
11     "title": "To Kill a Mockingbird",
12     "category": "Fiction"
13  },
14  {
15     "_id": "6a314b426d1d142eca1ec0b3",
16     "totalBorrows": 3,
17     "title": "1984",
18     "category": "Fiction"
19  }
20  ]
```


Test Fines

POST

http://localhost:4000/api/fines/calculate

Params

Authorization

Headers

Body

Pre-request Script

Tests

Settings

1

{

2

"transactionId": "6a314e1854a831c61d0304ca"

3

}

Body

Cookies

Headers (7)

Test Results

Pretty

Raw

Preview

Visualize

JSON v

1

{

2

"transactionId": "6a314e1854a831c61d0304ca",

3

"memberName": "John Doe",

4

"bookTitle": "The Great Gatsby",

5

"dueDate": "2026-07-01T10:15:00.000Z",

6

"overdueDays": 3,

7

"finePerDay": 5,

8

"totalFine": 15

9

}

POST

http://localhost:4000/api/fines/pay

Params

Authorization

Headers

Body

Pre-request Script

Tests

Settings

1

{

2

"memberId": "6a314b426d1d142eca1ec0aa",

3

"amount": 15

4

}

Body

Cookies

Headers (7)

Test Results

Pretty

Raw

Preview

Visualize

JSON v

1

{

2

"message": "Fine paid successfully",

3

"memberId": "6a314b426d1d142eca1ec0aa",

4

"amountPaid": 15,

5

"remainingFine": 0

6

}

GET

http://localhost:4000/api/fines

Params

Authorization

Headers

Body

Pre-request Script

Tests

Settings

Body

Cookies

Headers (7)

Test Results

Pretty

Raw

Preview

Visualize

JSON v

```
1  [  
2  {  
3    "_id": "6a314f1254a831c61d0304cb",  
4    "member": {  
5      "name": "Jane Smith",  
6      "email": "jane@gmail.com"  
7    },  
8    "fineAmount": 50,  
9    "status": "Unpaid"  
10 }  
11 ]
```